

A Rational Two-Point Counterexample to the Zero-Radical Clause for TLMC 00000002617

Prepared for the account `idealistic hacker`

Frozen statement and scope

The organizer source was frozen before this analysis:

Source: `conjectures/00000002617.md`
Commit: `95acb520ec5607c826b8a997b1ef2fc82d6f7c57`
Git blob: `e23e80d92bfbf3e0970c427716f7ce3c2f97a39e`
SHA-256: `03ba0f9febb435b53de0c4df73642bbb647b1a0b9eef3f297cedf5263564bc5b`

We address only the zero-radical clause at the conventional, fixed instance

$$A = \text{IncidenceAlgebra}(\mathbb{Q}, \text{Fin } 2), \quad \text{Fin } 2 = \{0 < 1\}.$$

This is a finite locally finite chain with rational coefficients. We do not make a statement about arbitrary coefficient rings or arbitrary posets, and neither define nor use a notion of semisimplicity. The source's Möbius-function language is not a proof object in this package.

Calculation in the actual incidence algebra

Let $e \in A$ take the value 1 on the strict interval $(0, 1)$ and 0 elsewhere. This is an actual Mathlib `IncidenceAlgebra` function, not a substitute coordinate algebra. The Lean file `lean4/Main.lean` expands Mathlib's `IncidenceAlgebra.mul_apply` over all pairs in `Fin 2` and proves

$$(ye)^2 = 0 \quad \text{for every } y \in A.$$

It also proves $e \neq 0$.

Jacobson-radical bridge

Mathlib's actual `Ideal.mem_jacobson_iff` characterizes membership of x in the Jacobson radical of an ideal. For each y , choose $z = 1 - ye$. Since $(ye)^2 = 0$, its required expression simplifies to

$$z ye + z - 1 = (ye - (ye)^2) + (1 - ye) - 1 = 0.$$

Thus Lean proves

$$e \in \text{Ideal.jacobson}(\perp : \text{Ideal } A).$$

Since $e \neq 0$, it follows that

$$\text{Ideal.jacobson}(\perp : \text{Ideal } A) \neq \perp.$$

Hence the zero-radical clause has a counterexample under the explicitly fixed rational two-point interpretation.

Lean evidence and trust boundary

The project pins Lean 4.33.1 and Mathlib Git commit `0df444a360eaa60ab8c11dca51a86af692955474`. `Check.lean` prints the transitive axiom footprints for all advertised theorems. The current proofs use the standard Lean/Mathlib principles `propext`, `Classical.choice`, and `Quot.sound`; no new axiom, `sorry`, `admit`, `native_decide`, or unsafe evaluation is used.

Status

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